A)Terminal 1 — start POX

```
kishansaaai@Ubuntu:~/pox$ cd ~/pox
python3 pox.py log.level --DEBUG openflow.of_01 link_failure_recovery
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:link_failure_recovery:Link Failure Detection and Recovery Controller Started
INFO:link_failure_recovery:Link Failure Detection and Recovery module loaded.
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.12.3/Mar 23 2026 19:04:32)
DEBUG:core:Platform is Linux-6.17.0-20-generic-x86_64-with-glibc2.39
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
```

B)Terminal 2 – Start Topology

```
kishansaaai@Ubuntu:~/pox$ sudo python3 ~/link_failure_topo.py
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1 s2 s3
*** Adding links:
(10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (h1, s1) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (h2, s1) (10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (h3, s3) (10.00Mbit 5ms delay) (10
.00Mbit 5ms delay) (h4, s3) (100.00Mbit 2ms delay) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
(100.00Mbit 2ms delay) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
(s1, s2) (10.00Mbit 20ms delay) (10.00Mbit 20ms delay) (s1, s3) (100.00Mbit 2ms delay) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
(100.00Mbit 2ms delay) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
(s2, s3)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...(10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (100.00Mbit 2ms delay) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
(10.00Mbit 20ms delay) (100.00Mbit 2ms delay) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
(100.00Mbit 2ms delay) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.
(10.00Mbit 5ms delay) (10.00Mbit 5ms delay) (10.00Mbit 20ms delay) (100.00Mbit 2ms delay) *** Error: Warning: sch_htb: quantum of class 50001 is big. Consider r2q change.


*** Topology started
Hosts: h1(10.0.0.1), h2(10.0.0.2), h3(10.0.0.3), h4(10.0.0.4)
Switches: s1 -- s2 -- s3 with backup link s1--s3
```

C)Automated Scenarios

```
=== SCENARIO 1: Normal Operation ===
Running ping h1 -> h3...
*** Ping: testing ping reachability
h1 -> X X X
h2 -> X X X
h3 -> X X X
h4 -> X X X
*** Results: 100% dropped (0/12 received)

=== SCENARIO 2: Simulating Link Failure (s1-s2) ===
Taking down link between s1 and s2...
Pinging after failure - controller should recover via s1-s3-s2...
*** Ping: testing ping reachability
h1 -> h2 X h4
h2 -> X X h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 25% dropped (9/12 received)

=== SCENARIO 3: Link Restoration ===
Link restored. Pinging again...
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
```

D)Manual Testing

Screenshot 1 - learn paths

mininet> pingall

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> X X X
h2 -> X X X
h3 -> X X X
h4 -> X X X
*** Results: 100% dropped (0/12 received)
```

Screenshot 2 - confirm 0% loss

mininet> pingall

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
```

Screenshot 3 - flow tables

mininet> dpctl dump-flows

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> X X X
h2 -> X X X
h3 -> X X X
h4 -> X X X
*** Results: 100% dropped (0/12 received)
mininet> dpctl dump-flows
*** s1 ------------------------------------------------------------------------
 cookie=0x0, duration=14.076s, table=0, n_packets=535, n_bytes=52430, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s1-eth4",vlan_tci=0x0000,dl_src=00:00:00:00:00:04,d
l_dst=00:00:00:00:00:02,nw_src=10.0.0.4,nw_dst=10.0.0.2,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth3"
 cookie=0x0, duration=8.959s, table=0, n_packets=4, n_bytes=168, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s1-eth4",vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=
00:00:00:00:00:02,arp_spa=10.0.0.4,arp_tpa=10.0.0.2,arp_op=1 actions=output:"s1-eth3"
 cookie=0x0, duration=8.958s, table=0, n_packets=4, n_bytes=168, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s1-eth4",vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=
00:00:00:00:00:02,arp_spa=10.0.0.3,arp_tpa=10.0.0.2,arp_op=1 actions=output:"s1-eth3"
*** s2 ------------------------------------------------------------------------
 cookie=0x0, duration=14.071s, table=0, n_packets=535, n_bytes=52430, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s2-eth1",vlan_tci=0x0000,dl_src=00:00:00:00:00:04,d
l_dst=00:00:00:00:00:02,nw_src=10.0.0.4,nw_dst=10.0.0.2,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s2-eth2"
 cookie=0x0, duration=8.963s, table=0, n_packets=4, n_bytes=168, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s2-eth1",vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=
00:00:00:00:00:02,arp_spa=10.0.0.4,arp_tpa=10.0.0.2,arp_op=1 actions=output:"s2-eth2"
 cookie=0x0, duration=8.962s, table=0, n_packets=4, n_bytes=168, idle_timeout=10, hard_timeout=30, priority=65535,arp,in_port="s2-eth1",vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=
00:00:00:00:00:02,arp_spa=10.0.0.3,arp_tpa=10.0.0.2,arp_op=1 actions=output:"s2-eth2"
*** s3 ------------------------------------------------------------------------
 cookie=0x0, duration=14.103s, table=0, n_packets=535, n_bytes=52430, idle_timeout=10, hard_timeout=30, priority=65535,icmp,in_port="s3-eth3",vlan_tci=0x0000,dl_src=00:00:00:00:00:04,d
l_dst=00:00:00:00:00:02,nw_src=10.0.0.4,nw_dst=10.0.0.2,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s3-eth4"
 cookie=0x0, duration=8.988s, table=0, n_packets=0, n_bytes=0, idle_timeout=10, hard_timeout=10, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:04,dl_dst=00:00:00:00:00:02,arp_spa=10.0.0.4,
arp_tpa=10.0.0.2,arp_op=1 actions=drop
 cookie=0x0, duration=8.980s, table=0, n_packets=0, n_bytes=0, idle_timeout=10, hard_timeout=10, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:02,arp_spa=10.0.0.3,
arp_tpa=10.0.0.2,arp_op=1 actions=drop
mininet>
```

Screenshot 4 - simulate failure and ping during failure (some loss expected)

mininet> link s1 s2 down

mininet> pingall

```
mininet> link s1 s2 down
mininet> pingall
*** Ping: testing ping reachability
h1 -> X X X
h2 -> h1 X X
h3 -> h1 h2 X
h4 -> h1 h2 h3
*** Results: 50% dropped (6/12 received)
```

Screenshot 5 - restore link - ping after recovery (0% loss again)

mininet> link s1 s2 up

mininet> pingall

```
mininet> link s1 s2 up
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
mininet>
```